Research on Chemical Intermediates
https://doi.org/10.1007/s11164-025-05843-7

Coriander powder-derived activated carbon (CPAC)-supported CuO nanocomposite for high-performance supercapacitor and catalytic applications

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Received: 8 September 2025 / Accepted: 23 November 2025
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Abstract

A cost-effective and sustainable coriander powder-derived activated carbon (CPAC) was synthesized directly from coriander powder using potassium carbonate without pre-carbonization. Copper oxide (CuO) nanoparticles were supported on this bio-derived CPAC matrix using a simple impregnation method to enhance the specific capacitance and catalytic performance, such as nitrophenol to aminophenol. The synthesized material was systematically characterized through FTIR, Raman, XRD, SEM, and SEM–EDX. The resulting CPAC@CuO nanocomposite exhibited excellent electrochemical performance, delivering a specific capacitance of 1241 F/g at 1.5 A/g in 3 M KOH electrolyte. The electrode upheld 61.7% capacitance retention after 5000 charge–discharge cycles with a coulombic efficiency of 99.7%, indicating superior cycle stability.

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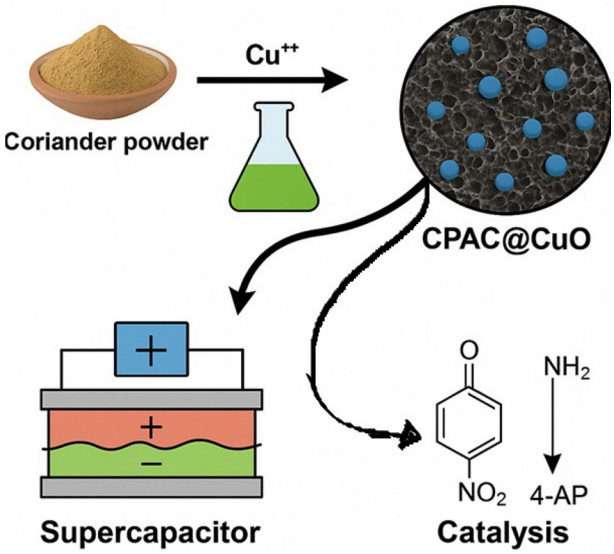
Extended author information available on the last page of the article

Published online: 24 December 2025

Graphical abstract



Keywords Coriander powder-derived activated carbon (CPAC) · Copper oxide nanoparticles (CuO) · Super capacitor · Catalytic reduction

Introduction

Materials or supported materials containing activated carbon have been utilized for many years as adsorbents, catalysts [1]. The highly developed pore architectures and large surface areas are versatile qualities of activated carbons [2, 3]. In addition, materials without activated carbon possess lower surface areas and fewer pores, and a smaller number of reactive sites, resulting in lower activity compared with activated carbons [4]. Hence, it is essential to enhance their activity to improve their surface areas, pore structures, and reactive sites. Nowadays, highly striking biomass-derived activated carbon has been obtained from farm and forestry waste owing to its easy sustainability and abundance as sources [5]. For this reason, several researchers are interested in creating carbon based on biomass that has a controlled carbonaceous structure [3]. The carbon materials' predecessor is determined by their price, purity, and availability. The production of activated carbon is reliant on carbon-rich organic materials, including wood, coal, porous polymers, agricultural waste, and vegetable and fruit waste [6–9]. There are numerous bio-waste products accessible, including lemon peels, grain sorghum, coconut shells, peanut shells, chestnut wood, and cilantro leaves, which are appropriate carbon materials for diverse applications [2, 3, 8, 10, 11]. Owing to this, the biowaste-derived activated carbons play a significant role in various applications such as energy storage, gas adsorption, and organic

pollutants, which are important for environmental protection. Therefore, coriander powder was used as a biosource for preparing carbon material, as it is widely known for its antioxidants, antidiabetic, metabolic syndrome and anti-inflammatory activity.

Generally, there are two methods for the preparation of activated carbon; those are physical and chemical activation. The physical activation has two stages, whereas chemical activation also has 2 stages: (i) the initial carbonization around 300–600 °C, followed by the carbon material activating with various activating reagents at 600–1000 °C; (ii) the raw material is directly mixed with activating agents and ground well with a mortar pestle or impregnation, followed by heating the material around 600–900 °C. Chemical activation is superior to physical activation because it may be carried out at a lower activation temperature and yields a higher amount of materials. Potassium salts such as KOH, K_2CO_3, $KMnO_4$, $KHCO_3$, KCl, and KBr are available for the activation of chemical compounds. Among these, the most frequently used KOH gives the highest surface area and highly porous material, but it is expensive and a toxic material compared with K_2CO_3 [12]. Very recently, AC Bose et al. [13] have used various potassium-based salts except for $KHCO_3$ for the activation of coconut tree husk for enhancing surface area and super capacitor application. He found that $KMnO_4$ gives a high surface area compared with KOH and K_2CO_3. Shaojun Yuan et al. [14] have used $KMnO_4$ as an activating agent for spinach waste for the removal of heavy metal ions [15]. In this current study, activated carbon was prepared from coriander powder using K_2CO_3 as the activating agent. The use of K_2CO_3 enhances carbon yield by releasing CO_2 gas during carbonization. Under humid conditions, using KOH tends to form a wet solid, which can lead to material loss during transfer to the ceramic boat and requires strong acid washing for neutralization.

Nanomaterials (NMs) play a vital role in various applications such as sensing, supercapacitors, catalysis, biology, and agriculture etc. [16, 17]. NMs have quantum and surface effects that enhance the physicochemical properties, such as mechanical, thermal, magnetic, catalytic, and optical properties. Numerous methods have been used for the preparation of NMs of valuable metals such as Ru, Pd, Rh, and Pt, but these metals are highly moisture sensitive, non-recyclable, and limited in supply. It is most important to develop a new method that induces a possibly high reduction of nitro compounds to amines using cost-effective metals like Cu, Fe, Ni, and Co [18]. Numerous methods have been developed for the conversion of nitro to amino reduction. However, due to certain constraints, like issues with selectivity, environmentally harmful impurities or byproducts were produced. Toxic or expensive reducing chemicals are used in some literature [19]. In the research industries or laboratories, the transformation of nitro compounds has received the most interest since amines are utilized as intermediates in the agrochemical, pigment, and pharmaceutical industries. Furthermore, it extended to apply for the reduction of methyl violet (MV), erochrome-black-t (EBT), and methylene blue (MB) dyes in polluted water. On account of the global freshwater problem being exacerbated by industrialization, climate change, and population growth, all of which have a substantial effect on water quality. Because most industries use them as coloring agents and dyes, pollution is increasing worldwide as a result of improper disposal of hazardous waste from a range of industries, including the textile, paint and varnish, food, cosmetics,

leather, pharmaceutical, paper, and pulp sectors; they used various cationic and anionic azo dyes widely [20].

Energy storage is crucial for fulfilling human needs both economically and globally. Supercapacitors' remarkable qualities, such as their high power and energy densities, extraordinary crystallinity, high porosity, long cycle life, excellent chemical and thermal stability, variety of frameworks, and large specific surface area, make them a promising option for energy storage applications [21]. Energy sources are divided into two categories based on their supply and recyclability: non-renewable and renewable. Non-renewable energy sources include fossil fuels like coal, oil, and natural gas, which are finite in supply. Renewable energy sources include geothermal energy, biomass-derived carbon, hydroelectric power, wind energy, solar energy, and recycled animal manure [22, 23]. Due to the significant importance of addressing water pollution and the energy crisis, there is a need to develop high-performance materials.

In this context, we reported a simple, sustainable, and cost-effective method for fabricating activated carbon from coriander biomass powder using potassium carbonate, followed by CuO nanoparticles' incorporation to produce a functional CPAC@CuO composite. This hybrid material was investigated for dual applications, such as an electrode material in high-performance supercapacitors and as a catalyst for the reduction of nitro compounds, specifically 4-NP to 4-AP. The electrochemical behavior and catalytic efficiency were systematically evaluated to highlight the energy transfer between the porous matrix and CuO nanoparticles. This study contributes to the development of multifunctional materials derived from renewable resources with potential for scalable application in energy storage and environmental remediation.

Experimental

Materials

Coriander powder (CP) was purchased from the local market in Gyeongsan, South Korea. Potassium carbonate (K_2CO_3), hydrochloric acid (HCl), copper nitrate trihydrate ($Cu(NO_3)_2.3H_2O$), N-methyl-2-pyrrolidone (NMP), and polyvinylpyrrolidone (PVP) were attained from Dae-Jung Chemicals, South Korea. The ultrapure water was obtained from a Milli-Q purification system (Millipore). All other reagents used in this study were of analytical grade and were utilized as supplied, without any additional purification.

Instrumentation and measurements

Fourier transform infrared (FTIR) spectroscopy was conducted using a Shimadzu IR Prestige-21 instrument at ambient temperature. The samples were prepared in the form of KBr pellets and analyzed across the spectral range of 4500–500 cm^{-1}. To assess the crystallinity of the electrode materials, an X-ray diffractometer equipped

with nickel-filtered Cu Kα radiation was used, scanning over a 2θ range from 10° to 80°. The surface morphology, particle size, and structural features of the synthesized materials were examined using scanning electron microscopy (SEM) with models S-300 and SU-700 from Hitachi.

Synthesis of activated carbon from coriander powder (CPAC)

Coriander powder and potassium carbonate (K_2CO_3) were combined in a 1:3 weight ratio and finely ground using a mortar and pestle to ensure uniform mixing. The prepared mixture was then placed in a ceramic crucible and subjected to pyrolysis in a tubular furnace at 700 °C for 1 hour under a continuous flow of nitrogen gas to maintain an inert atmosphere. After carbonization, the resulting material was soaked in a 2 M hydrochloric acid solution to eliminate residual K_2CO_3. The washing process continued until a neutral pH was reached, followed by thorough rinsing with deionized water. The final product was dried at 70 °C for 12 h to obtain coriander powder-derived activated carbon (CPAC). The yield of the activated carbon was approximately 50% relative to the original mass of coriander powder. This CPAC was subsequently used for electrochemical testing.

Synthesis of Cu(II)O Nanoparticles

The 0.1 M concentration of ($Cu(NO_3)_2.3H_2O$) was liquefied in 50 ml of deionized water; to this, a 0.1 M concentration of NaOH solution was gradually added under vigorous stirring until the pH reached 14 and the formation of black precipitates, was observed. It continued to stir for 1 h; after that the precipitate was washed with ethanol and deionized water until neutral pH was reached, and the precipitate was dried at 75 °C for 12 h. At last, the precipitate was calcined at 550 °C for 3 h.

Synthesis of CPAC@CuO composite

First, 10 g CPAC carbon was mixed with 150 mL of conc. HNO_3 in the beaker, and sonicated for 90 min. Thereafter, the activated carbon was filtered, and splashed with Millipore water until the pH reached 7. The washed material was transferred into an oven and dried for 24 h at 120 °C. Next, the acquired CPAC was mixed with 12 g of copper salt and 4.0 g of PVA dissolved in 250 mL of Millipore water. The mixed solution was continuously agitated for 30 min and then sonicated for another 15 min. After that, the required amount of aq. NH_4OH solution was added slowly to the above blend until the precipitate was obtained on AC. The mixture was then stirred further for 1 h at 60 °C, filtered, and cleaned with an excess volume of warm DI water. The finishing product was continuously desiccated for half a day at 70 °C.

Electrochemical measurements

To fabricate the working electrode, the CPAC@CuO composite was blended with polyvinylpyrrolidine (PVP) and N-methyl-2-pyrrolidone (NMP) in a mass ratio of

8:1:1. The resulting mixture was finely ground and uniformly coated onto a nickel foam substrate. The electrode was then dried at 60 °C for 12 h to ensure proper adhesion. Notably, no additional conductive additives such as acetylene black or polyvinylidene fluoride (PVDF) were incorporated into the electrode formulation. A three-electrode electrochemical system was employed for characterization using an Hg/HgO electrode as a reference and a platinum wire as the counter electrode. Electrochemical measurements were conducted in a 3 M KOH aqueous electrolyte. Cyclic voltammetry (CV) was performed across a potential window of -1.0 to 0.0 V at scan rates ranging from 5 to 100 mV/s. Additional analyses, including galvanostatic charge–discharge (GCD), electrochemical impedance spectroscopy (EIS), and long-term cycling stability assessments, were performed to comprehensively evaluate the electrochemical performance of the electrode.

Results and discussion

Characterization of CPAC and CPAC@CuO composite

The functional groups in CuO NPs, CPAC, and CPAC@CuO composite were determined through FTIR, and the spectra are presented in Fig. 1a. Figure 1a shows the typical absorption peaks of CuO NPs which appeared at 3369, 1366, and 609 cm^{-1} bands, indicating OH, C-O, and Cu–O matching with literature reports [24]. The main distinct peak of CuO appeared at609 cm^{-1}. The CPAC appeared at 2965, 2847, 2098, 1576, 1071, 954, 691 cm^{-1}. The FTIR peaks at 2922 cm^{-1} and 2852 cm^{-1} indicate that C–H stretching bonds. The FTIR bands at 1576 cm^{-1} correspond to the C = O bond stretching and vibration of the OH bond of –COOH existing on the CPAC surface. The peak at 954 indicates that -C = CH_2 in CPAC and a peak at 6642 cm^{-1} indicates the out-of-plane mode of C-H bending vibration after the aromatic ring. A comparison of the FTIR spectra of CuO, CPAC, and CPAC@CuO indicates the successful incorporation of CuO NPs into the CPAC matrix. Notably, a distinct peak at 609 cm^{-1}, characteristic of Cu–O stretching vibration, is present in both CuO and CPAC@CuO but absent in CPAC, confirming the presence of CuO in the composite.

In the Raman spectra (Fig. 1b), the pristine CPAC sample exhibits 2 well-defined bands at around 1350 cm^{-1} (D-band) and 1580 cm^{-1} (G-band), corresponding to disordered carbon (sp^3 hybridized) and graphitic carbon (sp^2 hybridized), respectively. The D/G intensity ratio provides insight into the defect density within the carbon structure.

In the CPAC@CuO NCs, the intensity of both D and G bands is substantially reduced, and the overall spectrum appears dampened. This reduction is indicative of CuO nanoparticles' coverage over CPAC, which restricts the Raman scattering from the carbon matrix. The lowered D/G ratio may also suggest reduced disorder or shielding of the CPAC surface due to CuO deposition.

The XRD patterns are shown in Fig. 1c. It reveals the structural difference between CPAC, and CPAC@CuO. It clearly shows that the CuO NPs were incorporated into activated carbon successfully. The activated carbon diffraction peaks

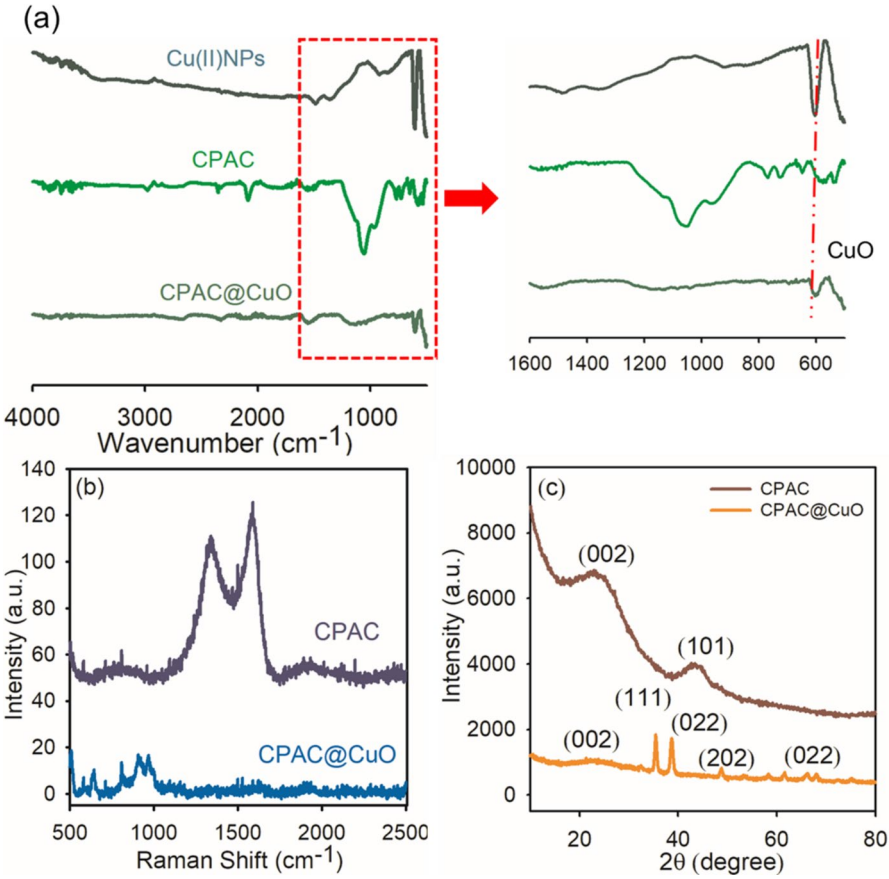


Fig. 1 **a** FT-IR, **b** Raman, and (**c**) XRD spectra of CuO NPs, CPAC, and CPAC@CuO

were observed at 20–30° at lattice plane (002), which represents that carbon is in an amorphous nature; whereas on the other hand, one more peak at 40–50° at lattice plane (101) indicates the graphitic carbon [25]. The existence of high-intensity peaks of CuO at 35.4° and 38.5° shows the copper oxide spreading on the surface of the CPAC [26]. The remaining XRD peaks at 48.5°, 58.4°, 62.1°, 66.2°, 68.3°, and 75.15° are clearly shown in the XRD spectrum. Furthermore, the XRD spectrum clearly shows that the CPAC carbon amorphous peak at 23.5° and the disappearance of the graphite carbon peak of CPAC at 43.6° indicate that the CuO NPs were incorporated into CPAC [27]. Moreover, the presence of CuO NPs crystalline peaks confirms the formation of CuO on the CPAC surface without altering its crystal structure. The composite also shows reduced intensity of the amorphous carbon peak, likely due to the CuO nanoparticles overlaying the carbon matrix.

The SEM images 2a and b depict the pristine CPAC structure. The material exhibits flake-like structures with a rough, irregular, and highly porous texture.

The pores highlighted in red in Fig. 2b were very clearly visible. These pores play a crucial role in hosting nanoparticles and enhancing surface-related interactions.

In contrast, images 2c and d reveal significant morphological changes after CuO nanoparticle loading. The porous structure is retained but now partially covered with the dispersed CuO nanoparticles, as indicated in the micrographs. These particles are mostly uniformly distributed. It suggests a successful impregnation of CuO on the CPAC surface. The presence of CuO is further confirmed by its distinct contrast and particular appearance, often visible as brighter regions. Moreover, the CuO deposition does not completely block the pores, indicating that the composite retains its porosity while gaining functionality from CuO. This morphology supports enhanced surface reactivity, making the CPAC@CuO composite a promising material for catalytic, adsorption, and energy storage applications.

Figure 3a displays the EDS elemental mapping of the CPAC@CuO composite, confirming the presence and distribution of key elements. The individual maps for carbon (C) oxygen (O), and copper (Cu) show a uniform dispersion across the particle surface. Carbon is attributed to the CPAC matrix, while the presence of oxygen and copper corresponds to the CuO nanoparticles anchored onto the porous carbon structure. The even distribution of Cu, and O suggests successful and homogeneous deposition of CuO, with minimal agglomeration.

Figure 3b presents the corresponding EDS spectrum, further validating the element composition. The strong C peak originates from the CPAC framework, whereas the O and Cu peaks confirm the presence of CuO. The absence of

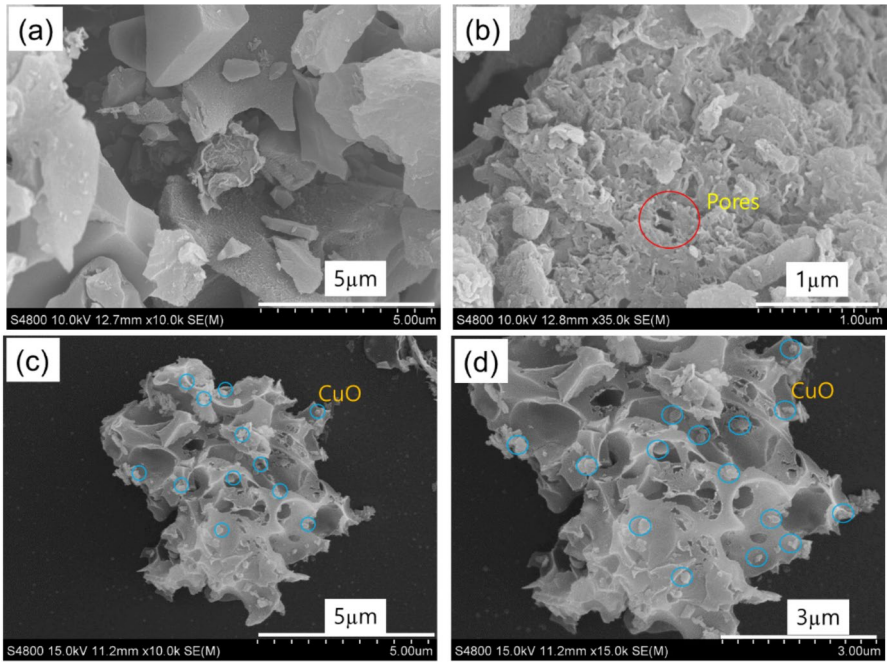


Fig. 2 **a** and **b** SEM images of CPAC; **c** and **d** SEM image of CPAC@CuO

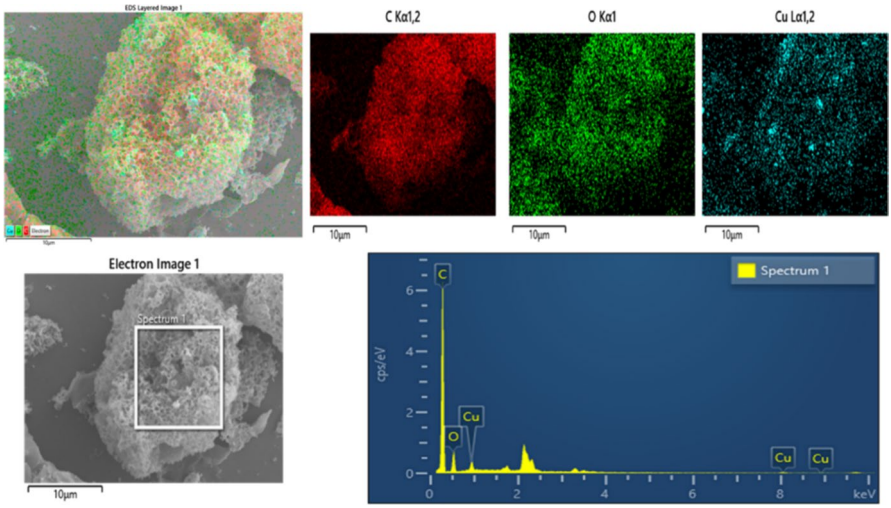


Fig. 3 **a** The EDS elemental mapping CPAC@CuO composite, **b** EDS spectrum of CPAC@CuO composite

additional impurity peaks indicates the high purity of the synthesized CPAC@CuO composite. Together, the elemental mapping and EDS spectrum provide strong evidence of successful CuO incorporation onto the CPAC matrix, in line with the SEM and XRD observations.

The nitrogen adsorption–desorption isotherms analysis, based on the Brunauer–Emmett–Teller (BET) method, is commonly used to determine the specific surface area, pore volume, and pore size distribution of synthesized material. A detailed BET analysis of CPAC@CuO is shown in Fig. 4. The isotherms correspond to a Type I curve, characteristic of microporous structures, exhibiting a

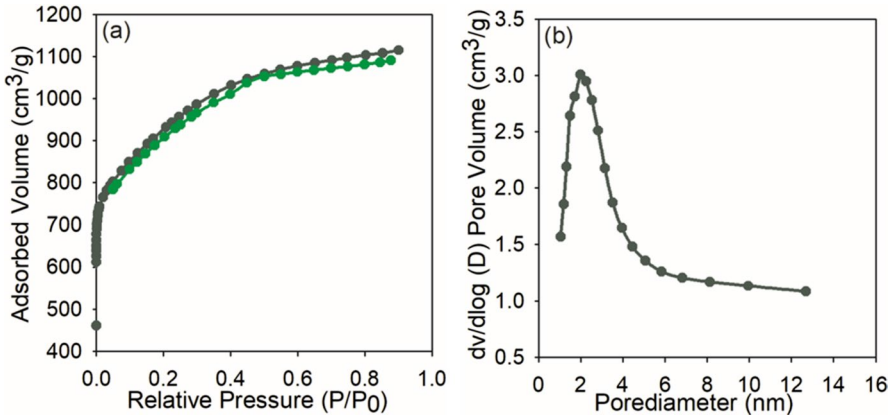


Fig. 4 **a** and **b** Surface area and pore size distribution of CPAC@ CuO

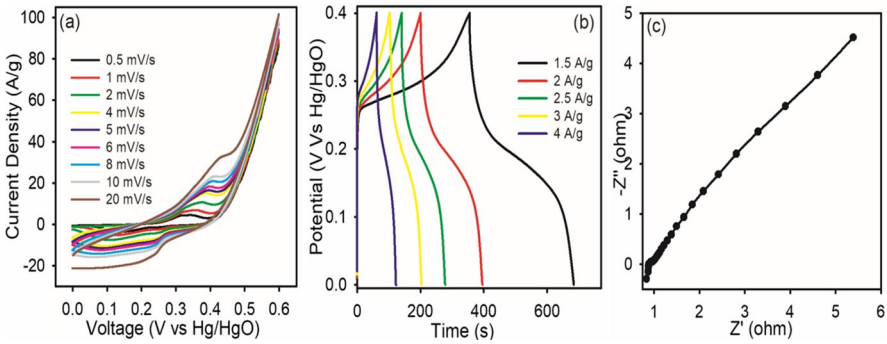


Fig. 5 **a** CV curves at various scan rates; **b** GCD at different current densities; **c** Nyquist plot of the prepared CPAC@CuO electrode

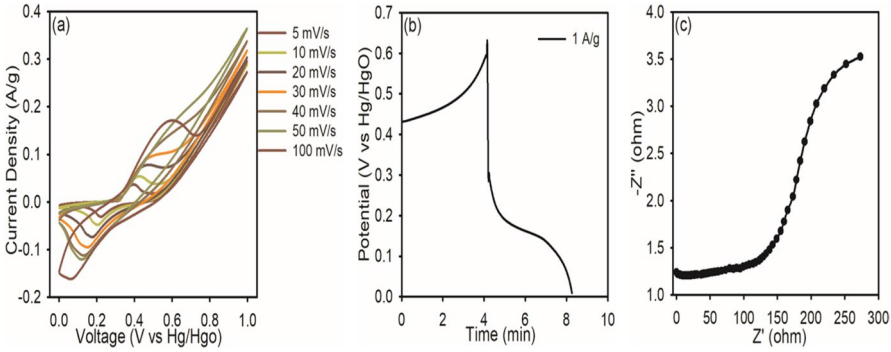


Fig. 6 **a** CV curves at different scan rates; **b** GCD at 1 A/g current density; **c** Nyquist plot of the prepared CPAC electrode

precipitous adsorption curve at low relative pressures because of their larger surface area of 1652 m^2/g, with a dominant size around 1.90 A^0 (Fig. 4.b).

Electrochemical properties

The electrochemical characteristics of the fortified electrodes were investigated using CV, GCD, and EIS. The obtained results are presented in Figs. 5 and 6. Figure 5a CV curves of CPAC showed oxidation at 0.4 to 0.5 V and reduction at 0.2 to 0.3 V peaks of Ni foam at various scan rates; there is no rectangular shape due to the potential variation. The CPAC@CuO electrodes' distinct oxidation and reduction peaks combine battery-type and pseudocapacitive mechanisms. One prominent oxide peak at 0.4 to 0.5 V was present in all CPAC@CuO curves and was linked to the conversion of Cu^+ to Cu^{2+}. Similarly, the convergence of Cu^{2+} to Cu^+ was attributed to the reduction peak at 0.2 to 0.3. According to CV curves, Cu-AC-x's energy storage mechanism combines battery-type capacitance and pseudo. Figure 6a displays the CPAC@CuO curves at different scan rates. It is evident that the redox peaks steadily faded as the scan rate

increased. The ions in the electrolyte may diffuse to the accessible pores at reduced scan rates, improving the insertion and extraction of the ions and the redox process [28]. The redox peaks' gradual shift was associated with the electrode material's growing charge diffusion polarization. Furthermore, even at 20 mV/s, the CPAC@CuO CV curves maintained a similar curve shape, indicating tremendous capacitive performance and quick ion transmission at the electrode interface. These characteristics may be connected to the hierarchical porous structure and short diffusion paths of CPAC.

The super-capacitive characteristics are further assessed using GCD plots of AC and CPAC@CuO composite electrodes at 1 and 1.5 A/g (Fig. 5b and 6b). The properties of the pseudo-capacitive were the reason why the GCD profile of AC was about nonlinear, with a high infrared drop at 1 A/g. Whereas in CPAC@CuO, one change in the slant of the potential was shown, which indicated that the reduction reaction happened between electrolyte ions and Cu/CuO NPs. For supercapacitors, specific capacitance is an important characteristic that changes with current density. The nonlinear GCD profiles of CPAC@CuO electrodes at various current densities displayed in Fig. 5b illustrate the interaction between the electrode and electrolyte through the faradaic reaction process. As the current density increases, the specific capacitance falls from 1241 to 630 A/g, as shown in Fig. 5b. The current densities range from 1.5 to 4 A/g. As the current density increased, the electrolyte ions' diffusion depth in the electrode material's pores shrank, which reduced the number of active sites. However, CPAC exhibits an insignificant specific capacitance of 10 A/g, as shown in Fig. 5b. For this reason, we did not continue studying it at different current densities. This pattern highlights the rate limits of the electrode material because the efficiency of the electrode material diminishes with increasing charging and discharging rates.

Table 1. Assessment of the specific capacitances of the CPAC@CuO electroactive material to the recently stated Cu-based carbon composite

Entry	Material	Electrolyte	Current Density (A/g)	Capacitance (F/g)	Refs
1	Cu-AC-2	6 M KOH	0.5	360	[28]
2	Cu–C	1 M KOH	1	318	[31]
3	CuS@ MOF-C	1 M H_2SO_4	0.5	321.9	[32]
4	CuO/GO	1 M Na_2SO_4	1	211	[33]
5	Fe_3O_4-Carbon composite	6 M KOH	5	8780	[34]
6	Wild mushrooms derived carbons	PVDF-HFP/$Mg(ClO_4)_2$	-	247	[35]
7	Waste tea leaves-derived activated carbon	1 M Na_2SO_4	1	250	[36]
8	Cassia fistula dry fruits	1 M KOH	1	583	[37]
9	pCBz@C	3 M KOH	0.5	215	[38]
10	Corncob-derived activated carbon	6 M KOH	1	271	[39]
11	CPAC@CuO	3 M KOH	1.5	1241	This work

More details regarding the interfacial dynamics, diffusion processes, electronic conductivity, and charge transfer resistance were supplied by EIS in the electrode-electrolyte system. The CPAC@CuO and CPAC electrodes' Nyquist plots are displayed in Figs. 5c and 6c. The displayed intercept on the real axis represents the solution resistance (Rs), and the diameter of the semicircle represents the charge transfer resistance (Rct). Interestingly, CPAC@CuO displayed a minimal Rs in comparison to the CPAC electrode, indicating that it is appropriate for electrochemical storage. Furthermore, CPAC@CuO has an Rct of 15.72 and a value of Rs 0.819 Ω, which demonstrates the insulated carbon's exceptional electrical conductivity. The remarkable rate capability was greatly enhanced by the low Rct. The straight-line slope, which indicated good capacitive behavior, demonstrated the rapid ion diffusion of electrolyte ions in the pores in the low-frequency zone [29]. The linear portion of CPAC@CuO was more vertical than that of CPAC, suggesting an optimal capacitor. Lower charge transfer resistance at the electrode/electrolyte interface was suggested by the semicircle's diameter in the high-middle frequency [30].

Comparing the CPAC and CPAC@CuO electrodes' electrochemical performance

Compares the CV curves of the two electrodes at a scan rate of 20 mV/s. In comparison with CPAC, the bigger area of CPAC@CuO indicates a higher capacitance. Additionally, the longer discharge time in comparison to CPAC and consequently better capacity to store and release charge are indicated by the GCD curves at 1.5 A/g current density. Because of its unique porous structure, which is further strengthened by CuO NPs, it allows for chemical interaction between the electrolyte and the electrode.

Table 1 demonstrates the comparison between the acquired results and the reported copper oxide and carbon composite materials. Cu-modified activated carbons generated from biomass were reported by the L.I. Shuang [28] group in 2022; they demonstrated 360 F/g at 0.5 A/g. In 2020, the W-G Zhang [31] group synthesized a Cu-MOF-derived Cu–C nanocomposite that demonstrated 318 F/g at 1 A/g. The electrochemical characteristics of copper sulfide (CuS) plated on metalorganic

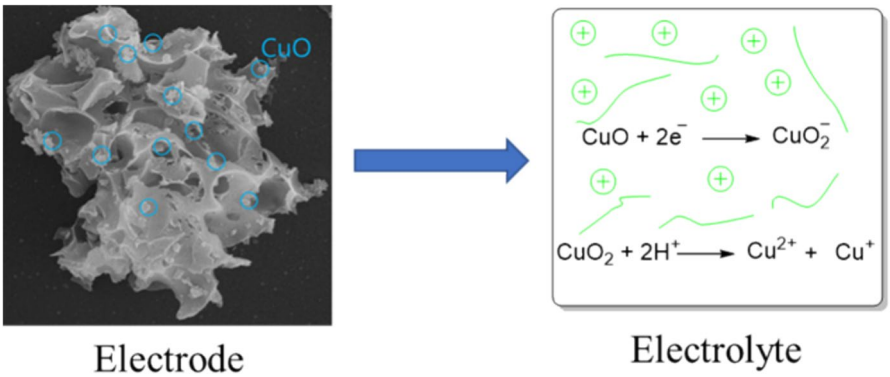


Fig. 7 Mechanistic representation of the interaction between electrode and electrolyte

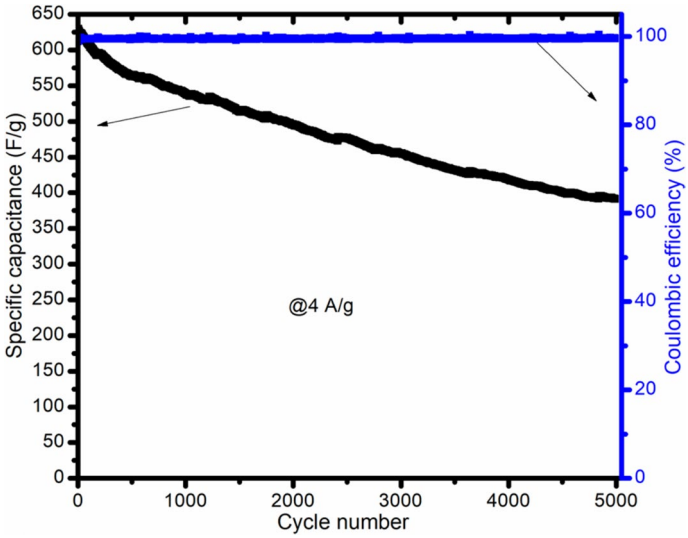


Fig. 8 Cycling stability of CPAC@CuO over 5000 cycles

framework (MOF) generated carbon were investigated by Y. Liu et al. [32] and revealed 321.9 F/g at 0.5 A/g. According to research by W. Liu et al. [33], the copper oxide/graphene oxide (CuO/GO) aerogel composite demonstrated 211 F/g at 1 A/g. M. Saxena et al. [34] have fabricated an Fe_3O_4-carbon composite, which was evaluated for a supercapacitor application and exhibited high specific capacitance due to iron oxide. Similarly, A. Jain et al. [35] reported the synthesis of carbon material derived from wild mushrooms for use in both super capacitor and dye adsorption studies; the carbon material demonstrated performance even without incorporating metal ions. F. Khan et al. [36] synthesized activated carbon from waste tea leaves for super capacitor application, achieving a specific capacitance of 282 F/g at a current density of 0.9 A/g. Moreover, S-F Wang et al. [37] developed activated carbon material from *Cassia fistula* dry fruits at four different temperatures (i.e. 500, 600, 700 and 800 °C). Among these, the sample prepared at 800 °C exhibited the best performance, due to high surface area and pore diameter, which enhanced the interaction between the electrode and electrolyte. In 2025 J. Lee et al. [38] synthesized a carbazole-based porous carbon material for super capacitor and dye adsorption applications, showing a specific capacitance of 215 F/g at 0.5 A/g, though its adsorption capacity was relatively low. Compared to these previous reports, the synthesized CPAC@CuO composite showed significantly improved specific capacitance and offered simple fabrication. The superior performance of the present CPAC@CuO composite compared to other reported materials can be attributed to its unique porous carbon structure and the synergistic interaction between CuO nanoparticles and the activated carbon matrix. The chemical activation facilitates the development of abundant porous (micro, meso and macro) structures, enhancing the surface area and ion diffusion pathways. Additionally, the CuO nanoparticles anchored onto the

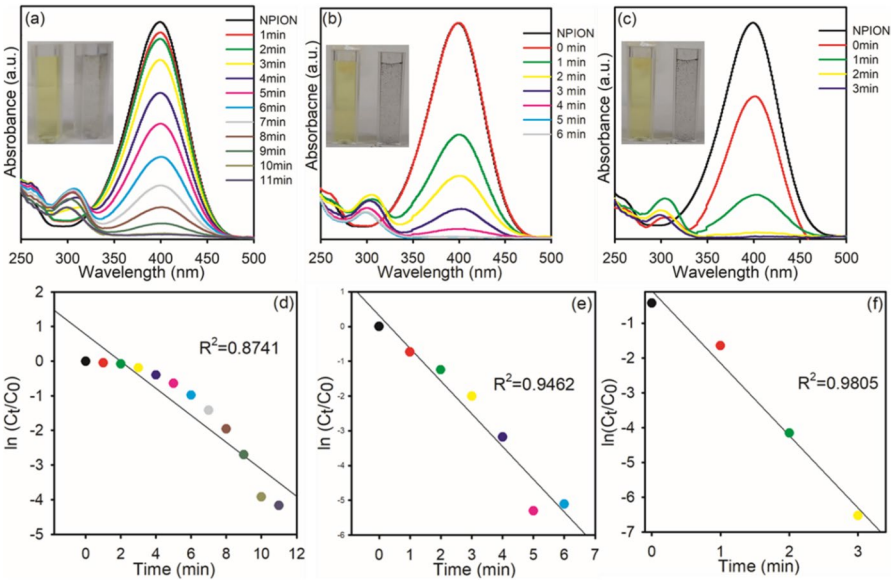


Fig. 9 **a**, **b**, and **c** Time-dependent UV–Vis spectra of the reduction of 4-NP in the presence of CPAC@CuO (1.0 mg, 2.5 mg, and 5.0 mg); **d**, **e**, and **f** Kinetic study of catalytic reduction

carbon surface acted as active redox centers, improving electrical conductivity and charge storage capability through reversible Cu^{2+}/Cu^{+} redox transitions (Fig. 7).

Figure 8 illustrates the long-term cycling performance of the CPAC@CuO composite electrode, evaluated over 5000 consecutive GCD cycles. Throughout the

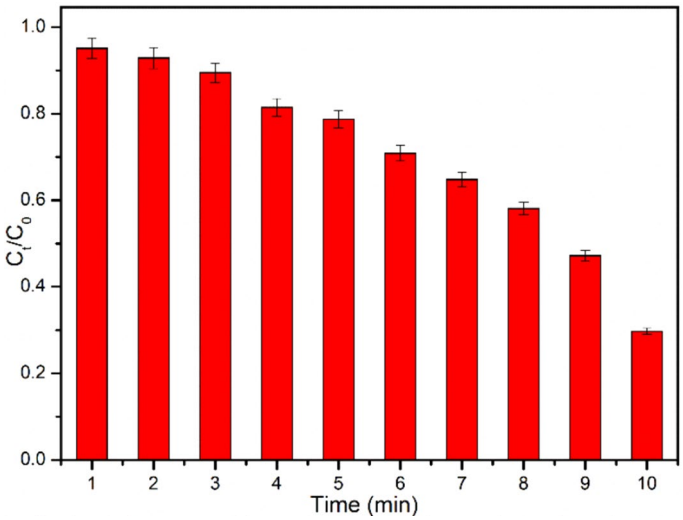


Fig. 10 Error bar diagram for catalytic reaction at different time intervals

entire cycling test, the specific capacitance remains consistently stable, indicating no noticeable decline in electrochemical performance. Impressively, the electrode maintains 61.7% of its original capacitance and achieves a high coulombic efficiency of 99.7% after 5000 cycles. This outstanding cycling stability highlights the robust structural integrity and excellent electrochemical durability of the CPAC@CuO material. The results suggest that the combination of activated carbon with CuO creates a synergistic effect, where activated carbon enhances electrical conductivity and CuO contributes pseudo-capacitive behavior. This synergy ensures both high capacitance and strong resistance to degradation during repeated cycling, making the CPAC@CuO electrode a strong candidate for use in reliable and long-lasting supercapacitor devices.

Catalytic reduction of nitrophenol to aminophenol

The model reaction utilized to assess the catalytic activity of the produced CPAC@CuO is the catalytic reduction of PNP to PAP in the presence of $NaBH_4$. This was achieved by combining 5 mg of CPAC@CuO, 0.15 mL (0.01 mmol) of $NaBH_4$, and 1.85 mL (1 mmol) of 4-NP (the model pollutant) in a UV cuvette. The reaction's progress was monitored at certain intervals. 4-NP shows an absorption peak at 317 nm in response to this $NaBH_4$ addition. The OH group of 4-NP deprotonates and generates the 4-nitrophenolate ion, causing the absorption peak to shift from 317 to 400 nm. Additionally, the light-yellow color of 4-NP transforms into a deep yellow color. [40]. Due to the rapid migration of hydride ions from $NaBH_4$, metal nanoparticles can improve the reduction of 4-NP to 4-AP. This is because literature findings have shown that the hydride ions of $NaBH_4$ and 4-nitrophenolate ions are repulsed by one another. The nitrophenolate ion peak strength at 400 nm rapidly declined as the reaction went on, and a new peak at 300 nm formed, which amply demonstrated the creation of PAP. Additionally, it is confirmed that PAP forms at the isosbestic sites of 281 and 314 nm without any byproducts, and the solution's deep yellow color turns colorless. Furthermore, we have studied the effect of catalyst amount for the reduction reaction, and the obtained results are presented in Fig. 9. Figure 9 reveals that if the amount of catalyst is increased, the rate of reaction is also increased due to the interaction between the ions of CuO and $NaBH_4$. When the catalyst amount of 1 mg was used in the reaction, it took 11 min to complete the reaction. When the catalyst amount was increased from 1 to 2.5 mg, the reaction completed in 6 min. When the amount of catalyst was increased to 5 mg, the reaction was completed in 3 min. Based on this result, the amount of catalyst is enhancing the rate of reaction. Additionally, the reduction process's kinetics were examined using spectrophotometry, as illustrated in Fig. 9(d, e, f), which confirms that the catalytic reduction follows pseudo-first order with a rate constant of 0.8741, 0.9462, and 0.9805 min^{-1} and the error bar diagram was provided in (Fig. 10).

Conclusion

In this study, we developed a sustainable and cost-effective nanocomposite (CAPC@CuO), synthesized using coriander powder-derived activated carbon (CPAC) and CuO nanoparticles. The synthesis approach employed potassium carbonate as the activating agent, eliminating the need for pre-carbonization and toxic chemicals. This green strategy provides an efficient route for converting agricultural biomass into functional materials for advanced applications. Electrochemical investigations revealed that CPAC@CuO exhibits excellent capacitive performance with a specific capacitance of 1241 F/g at 1.5 A/g in 3 M KOH electrolyte. The material also demonstrated impressive long-term cyclic stability, retaining 61.7% of its initial capacitance after 5000 charge–discharge cycles and maintaining a coulombic efficiency of 99.7%, highlighting its potential as a durable electrode for supercapacitor devices. The superior performance is attributed to the hierarchical porous structure and uniform dispersion of CuO nanoparticles, which facilitated fast ion diffusion and reversible redox reactions. It indicates superior cycle stability and an effective electrode material for the next generation of supercapacitors with high power and great dependability. This study suggested a novel approach to utilizing high-energy capacitive materials in a variety of amorphous materials, which could facilitate the deployment of large-capacity and high-rate energy storage in practice. Beyond energy storage, CPAC@CuO also showed outstanding catalytic activity for the conversion of para-nitrophenol to para-aminophenol in the presence of $NaBH_4$ within 3 min using only 5 mg of catalyst, following pseudo-first-order kinetics. This rapid, effective reduction highlighted its practical potential in wastewater treatment processes.

Acknowledgements The author Sivarama Krishna Lakkaboyana thanks Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology for providing the RDF-Seed Money and Seed Fund (Sanction order No: 021). The author, Salah Knani, extends their appreciation to the Deanship of Scientific Research at Northern Border University, Arar, KSA for funding this research work through the project number “NBU-FFR-2025-2483-05".

Author contributions SRKL and RMNK: Writing—original draft, Methodology, Investigation, Formal analysis, Data curation. SVP and JL: Writing—original draft, Validation, SP, SK and RA: Review & editing.

Funding No source of funding exists for this manuscript.

Data availability The data provided for the results presented in this study are available through the corresponding author upon request.

Declarations

Competing interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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